Day 5 Student Notes: Claude Code for Infrastructure Repositories

> **Today in one line:** Claude Code works inside a code repository. It reads many files, plans, edits and reports. You stay in control with plans, diffs and rules.

What is Claude Code?

Same Claude, but with tools. It reads your whole repo, plans changes, edits several files together, can run commands and read the output, and works with Git. It loops: **Gather** information, **Act**, **Verify**. You can stop or steer it at any time.

Cowork or Claude Code?

	Cowork	Claude Code
Built for	Documents, reports, simple files	Code and infrastructure in a Git repo
Runs commands?	Focus is files	Yes: tests, linters, scripts, Git
Fix loop	You check the result	It runs a check, reads the error, fixes it, re-runs
DevOps example	Write the postmortem	Fix a Terraform file and commit on a branch

Setup (do this once)

You need: Claude Pro (or Max, Team, Enterprise), VS Code version 1.94 or newer, the sample repo (shopeasy-infra.zip) unzipped to a folder you can find. On Windows, Git for Windows is recommended. No API key: you sign in with your Claude account.

Step 1: Install the extension in VS Code

1. Open Extensions: Ctrl+Shift+X (Mac: Cmd+Shift+X).
2. Search **Claude Code**. Choose the one by **Anthropic**. Click **Install**.
3. Click the **Spark icon** (top right of the editor, or the left sidebar). The editor icon only shows while a file is open.
4. Click **Sign in**. Your browser opens. Approve. Return to VS Code.

If you do not see it: restart VS Code, or open the Command Palette (Ctrl+Shift+P) and run **Developer: Reload Window**.

Step 2 (optional): the terminal version

The extension works on its own. Install this only if you want to type **claude** in a terminal.

```
Windows PowerShell:  irm https://claude.ai/install.ps1 | iex
Mac or Linux:        curl -fsSL https://claude.ai/install.sh | bash
```

Then open a **new** terminal and run **claude --version**. If it says "not recognized", close and reopen the terminal.

Step 3: Connect to the repo

1. VS Code: **File, Open Folder**. Choose **shopeasy-infra**.
2. Click **Yes, I trust the authors**.
3. Open the Claude panel with the Spark icon.

Claude works in the folder you opened. Always open the repo folder itself, never your whole Documents or Desktop.

How Claude Code behaves

1. **Plan:** it explores and tells you what it will do. Read the plan before approving.
2. **Edit:** it changes one or many files. You see a side-by-side diff with **Accept this change** and **Reject this change**.
3. **Report:** it tells you what changed and what to check.

Made a mistake? Press **Esc twice** to rewind file changes, or ask Claude to undo. Git also shows every change. Rewinding covers files only, not anything outside the folder.

Guardrails: the modes

Mode	What it does	Use for infra
Plan	Explores and proposes. No edits until you approve	Start here
Manual	Asks before edits and most commands	Normal work
Edit automatically	Edits files without asking	Only on practice files
Auto	A safety checker approves actions for you	Not on infra repos

Click the mode at the bottom of the prompt box to change it. Newer versions may start in Auto, so **check the mode before your first prompt**. In the terminal, Shift+Tab cycles modes.

Hands-on prompts (also in Day05_Prompts_To_Paste.txt)

1. Explore, change nothing (Plan mode)

```
Role: You are a careful senior DevOps engineer.
Context: This is a small practice repo for a made-up shop. It has Terraform,
Kubernetes, Docker, a shell script and a pipeline file.
Task: Explain the repo structure in simple words. Then flag ONE improvement that
matters most for security.
Constraints: Read only. Do not edit, create or delete any file. Do not run any
command.
Format: One line per folder, then the improvement with the file name.
```

Check: Did it list every folder? Does the improvement name a real file? Open the file and confirm.

2. Fix one thing, read the plan first

```
Task: Fix only the improvement you flagged. First show me your plan and wait.
Constraints: Change only that one file. Do not run any command. Add a short comment
that explains the change.
Format: After the edit, summarise what changed in 3 bullets.
```

Read the plan. Approve, or reply with changes. A diff opens. Check every changed line, then Accept or Reject.

3. Give Claude permanent rules

```
Create a CLAUDE.md file for this repo with these safety rules:
1. Never run terraform apply, terraform destroy or kubectl apply.
2. Always show a plan before editing any file.
3. Never print or invent secrets.
4. Explain each change in simple words.
```

CLAUDE.md is read at the start of every session, so your safety rules travel with the repo. Typing **/init** makes a starter one.

Safe-usage rules for infrastructure code

1. **Read first.** Start in Plan mode.
2. **Approve edits one by one.** Read every diff.
3. **Humans deploy.** No terraform apply, destroy or kubectl apply by the agent.
4. **Use Git and a branch.** You can always see and undo changes.
5. **No real secrets** in files or prompts.
6. **Test in a practice or non-production setup** first.

Useful commands

Command	What it does
/help	Show commands
/init	Create a starter CLAUDE.md
/plan	Plan mode
/clear	Start fresh conversation
/resume	Continue an earlier conversation
/doctor	Check your install
claude	Start in a terminal (CLI)
claude --version	Check the CLI is installed

Key words

Word	Meaning
Repository (repo)	A project folder tracked by Git
IaC	Infrastructure as code: Terraform and similar files
Diff	Side-by-side view of what changed
Plan mode	Claude proposes, you approve, nothing is edited first
CLAUDE.md	Rules file Claude reads every session
Agent	An AI that takes actions, not only writes text

Check yourself

1. Name the three steps of the Claude Code loop.
2. Which mode should you start in on an infra repo?
3. What does CLAUDE.md do?
4. Cowork or Claude Code: fix a Terraform file?
5. Name two rules for safe use.

Answers

1. Gather, act, verify (plan, edit, report is how you see it).
2. Plan mode.
3. Gives Claude permanent rules for the repo, read at the start of each session.
4. Claude Code.
5. Any two of the six rules.

Take-home task (20 to 30 minutes)

1. Open shopeasy-infra again (a fresh unzip is fine) and ask Claude Code to explain a different file.
2. Ask for a second improvement. Read the plan and the diff.
3. Bring one fix you accepted and one you rejected, with your reason. No real secrets or company repos.

> **Next class, Day 6:** Claude for Linux Administration: shell scripts and log analysis.